

CHAPTER 2

Activities & Lab Packet

The Sun, Earth & Moon

A complete set of labs, worksheets, writing prompts, discussions, and group projects to accompany Chapter 2. Works with both the Standard and Easy Read versions of the textbook.

 Georgia Standard S6E2

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1 Moon on a Stick — Modeling the Phases

LAB

OBJECTIVE

Model why the Moon shows phases and discover that the "shape" depends on our viewing angle.

TIME

30–40 minutes

STANDARDS

S6E2.a

Materials (per group of 3–4)

- 1 white foam ball (or ping-pong ball) pushed onto a pencil or skewer — this is the Moon
- 1 bright lamp with the shade removed, placed in the center of the room — this is the Sun
- A room that can be darkened
- Student data sheet (next page)

Setup

Place the lamp at head height in the middle of the room and darken everything else. Each student's **head is the Earth** — their eyes are the view from home. Spread groups out so shadows don't overlap. Walk through position 1 (ball toward lamp) together before letting groups continue alone.

Student Instructions

1. Stand facing the lamp. Hold your Moon at arm's length, slightly above your head, between you and the lamp. Look at the side facing YOU. How much is lit? This is the **new moon**.
2. Keep your arm out and slowly turn your body to the LEFT (counter-clockwise). Stop every one-eighth of a turn — each stop is one phase.
3. At each stop, sketch exactly what you see lit on the ball on your data sheet (positions 1–8).
4. At the halfway point (back to the lamp), the whole face glows — the **full moon**.
5. Answer the conclusion questions. Key question: did the ball ever actually change shape?

⚠ SAFETY & TEACHER NOTES

Safety: Bulbs get hot — no touching, and keep paper away from the lamp.

Common confusion: Students believe Earth's shadow causes phases. Ask mid-lab: "Is your head's shadow falling on the ball right now? So what's making the dark part?"

Tip: Holding the ball slightly above head level prevents accidental "eclipses" — then you can lower it on purpose later as a preview of Lab 2!

ANSWER KEY (data sheet)

Positions 1 → 8: new moon, waxing crescent, first quarter, waxing gibbous, full moon, waning gibbous, last quarter, waning crescent | 1) No — the ball stayed round; only our view of the lit half changed | 2) The lamp (Sun) always lit half the ball | 3) The half not facing the lamp — NOT a shadow from my head | Conclusion: phases come from seeing different amounts of the sunlit half as the Moon orbits.

Moon on a Stick — Data Sheet

Name: _____

Period: _____

Shade in the part of the ball you saw LIT at each position (1 = facing the lamp, then turning left):

Position 1	Position 2	Position 3	Position 4
			
Phase name:	Phase name:	Phase name:	Phase name:
Position 5	Position 6	Position 7	Position 8
			
Phase name:	Phase name:	Phase name:	Phase name:

1. Did the ball ever actually change shape? What changed instead?

2. How much of the ball did the lamp light up at every position?

3. What made the "dark part" of your Moon — your head's shadow, or something else? Explain.

Conclusion: Using the words ORBIT, SUNLIT HALF, and POINT OF VIEW, explain why the Moon has phases.

2 Shadow Play — Modeling Eclipses

LAB

OBJECTIVE

Model solar and lunar eclipses and explain why eclipses don't happen every month.

TIME

30 minutes

STANDARDS

S6E2.b

Materials (per group of 3–4)

- 1 lamp or strong flashlight (the Sun)
- 1 larger ball, e.g. tennis ball or globe (the Earth)
- 1 smaller ball on a skewer, e.g. ping-pong or foam ball (the Moon)
- Student data sheet (next page); a darkened room

Setup

One student holds the Earth still; another "orbits" the Moon around it while the lamp shines from one side. Remind students: in this model, sizes and distances are NOT to scale (the real Sun is enormous and very far away).

Student Instructions

1. **Solar eclipse:** slowly move the Moon between the lamp and the Earth until the Moon's shadow falls on the Earth ball. Freeze! Record the order of the three objects and where the shadow lands.
2. Look at the Earth ball: does the shadow cover the whole Earth or just a small spot? Record it.
3. **Lunar eclipse:** continue the Moon's orbit to the far side, so the Earth sits between the lamp and the Moon. Move until the Earth's shadow falls on the Moon. Freeze and record the new order.
4. **Why not monthly?** Now orbit the Moon again, but hold it slightly HIGHER than the Earth ball as it passes in front and behind. Watch where the shadows go. Record what happens.

⚠ SAFETY & TEACHER NOTES

Safety: Never look directly into the lamp/flashlight — and remind students the same rule applies to the real Sun during a real solar eclipse!

Key vocabulary move: Have students say the sentence "The _____ is in the middle, so its shadow falls on the _____" at each freeze point.

Common confusion: Students think eclipses and phases are the same thing. Step 4 is the antidote — the tilted orbit usually makes shadows MISS.

ANSWER KEY (data sheet)

1) Sun → Moon → Earth; shadow on Earth; only a small spot (only people in that spot see it) | 2) Sun → Earth → Moon; Earth's shadow darkens the Moon | 3) Solar = Sun blocked; lunar = Moon darkened; the middle object casts the shadow | 4) When the Moon is higher (tilted orbit), the shadows miss — that's why eclipses are rare, not monthly.

Shadow Play — Data Sheet

Name: _____

Period: _____

1. SOLAR ECLIPSE — Write the order of the three objects and sketch the setup, showing where the shadow falls:

Order: _____ → _____ → _____

Sketch (label Sun, Earth, Moon, and the shadow)

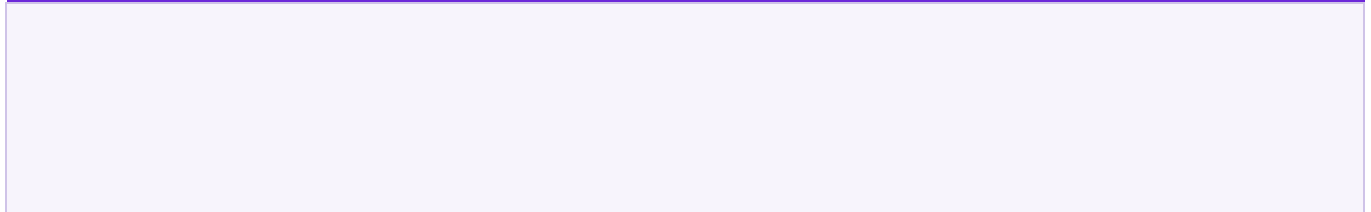


Did the shadow cover all of Earth or just a spot? What does that mean for who sees the eclipse?

2. LUNAR ECLIPSE — Write the new order and sketch it:

Order: _____ → _____ → _____

Sketch (label Sun, Earth, Moon, and the shadow)



3. Fill in the blanks: In a solar eclipse, the _____ gets blocked. In a lunar eclipse, the _____ gets darkened. The object in the **middle** is the one casting the _____.

4. When you held the Moon slightly higher as it orbited, what happened to the shadows? Why don't we get eclipses every month?

3 The Flashlight Test — Direct vs. Slanted Light

LAB

OBJECTIVE

Measure how the angle of sunlight changes how concentrated its energy is — the real cause of the seasons.

TIME

30 minutes

STANDARDS

S6E2.c

Materials (per group of 2–3)

- 1 flashlight (a phone light works)
- 1 sheet of graph paper or centimeter-grid paper, taped to the desk
- Ruler, pencil, and a book to keep the flashlight at a fixed height
- Student data sheet (next page)

Setup

The flashlight is the Sun; the graph paper is the ground. The trick to a fair test: keep the flashlight the **same distance** from the paper in both trials — only the **ANGLE** changes. (That's the whole point: same Sun, same distance, different angle.)

Student Instructions

1. **Trial 1 — Direct light (summer):** hold the flashlight about 20 cm directly above the paper, pointing straight down. Have a partner trace the lit circle.
2. Count the number of grid squares inside the traced shape. Record it.
3. **Trial 2 — Slanted light (winter):** keep the flashlight 20 cm away, but tilt it to about a 30° angle. Trace the new lit shape (it stretches into an oval).
4. Count the grid squares inside the oval. Record it.
5. The flashlight gave out the SAME light both times. Decide: in which trial did each square of ground get more energy? Complete the data sheet.

⚠ SAFETY & TEACHER NOTES

Tip: Darken the room — crisp light edges make tracing easier.

The payoff question: "Georgia gets slanted light in winter and direct light in summer. Earth never changes its distance much. So what causes the seasons?" Let THEM say "the tilt."

Extension: Tape the paper to a globe at the equator vs. near a pole and repeat — same idea, planet-shaped.

ANSWER KEY (data sheet)

1) The slanted beam covers MORE squares (bigger oval) | 2) Direct: small area = energy concentrated = stronger heating. Slanted: same energy spread thin = weaker heating | 3) Summer = direct (toward the Sun); winter = slanted (tilted away) | Conclusion: Earth's 23.5° tilt changes the ANGLE of sunlight on each hemisphere through the year — that, not distance, makes the seasons.

The Flashlight Test — Data Sheet

Name: _____

Period: _____

Trial	Flashlight angle	Shape of lit area	# of grid squares covered
1 — Direct (straight down)	90°		
2 — Slanted (tilted)	about 30°		

1. Which trial spread the light over MORE squares?

2. The flashlight gave off the same amount of light both times. In which trial did EACH square get more energy? Explain.

3. Match each trial to a season for YOUR hemisphere and explain the match:

Trial 1 (direct) = _____ because _____

Trial 2 (slanted) = _____ because _____

4. A friend says "summer happens because Earth gets closer to the Sun." Use your flashlight data to politely prove them wrong.

Conclusion: Using the words TILT, DIRECT, and SLANTED, explain the real cause of the seasons.

4 Phase Detective

WORKSHEET

OBJECTIVE

Identify, name, and order the eight phases of the Moon.

TIME

15–20 minutes

STANDARDS

S6E2.a

Name: _____

Period: _____

Part A. Number the phases 1–8 in order, starting at the new moon:

____ Full Moon

____ Waxing Gibbous

____ Waning Crescent

____ Last Quarter

____ New Moon

____ Waxing Crescent

____ First Quarter

____ Waning Gibbous

Part B. Fill in the blank with *waxing*, *waning*, *crescent*, or *gibbous*:

9. The lit part is getting bigger each night: the Moon is _____.

10. Less than half of the Moon is lit: it's a _____ moon.

11. More than half (but not all) is lit: it's a _____ moon.

12. We just passed the full moon, so now the Moon is _____.

Part C. Short answers:

13. Your little cousin says "the Moon is shaped like a banana tonight!" Explain what's really going on with the Moon's shape.

14. TRUE or FALSE: The dark part of the Moon is Earth's shadow. Explain your answer.

15. About how long does one full cycle of phases take? What motion causes it?

ANSWER KEY

A: New Moon 1, Waxing Crescent 2, First Quarter 3, Waxing Gibbous 4, Full Moon 5, Waning Gibbous 6, Last Quarter 7, Waning Crescent 8 | 9) waxing 10) crescent 11) gibbous 12) waning | 13) The Moon is always a round ball — we only see the part of its sunlit half facing us | 14) FALSE — the dark part is just the half not lit by the Sun; Earth's shadow only matters in a lunar eclipse | 15) About one month; the Moon orbiting Earth.

5 Solar or Lunar? — Eclipse Sort

WORKSHEET

OBJECTIVE

Distinguish solar and lunar eclipses by the order of the Sun, Earth, and Moon.

TIME

15 minutes

STANDARDS

S6E2.b

Name: _____

Period: _____

Part A. Write S (solar eclipse) or L (lunar eclipse):

____ 1. Order: Sun → Moon → Earth

____ 5. The daytime sky goes dark briefly

____ 2. Order: Sun → Earth → Moon

____ 6. The Moon turns a dim reddish color

____ 3. The Moon casts a shadow on Earth

____ 7. The SUN is the object being blocked

____ 4. Earth casts a shadow on the Moon

____ 8. The MOON is the object being darkened

Part B. Label each diagram SOLAR or LUNAR and draw where the shadow falls:

 Sun	 Moon	 Earth	 Sun	 Earth	 Moon
This is a _____ eclipse			This is a _____ eclipse		

Part C. Short answers:

9. What has to happen to the positions of the Sun, Earth, and Moon for ANY eclipse to occur?

10. The Moon orbits Earth every month. Why DON'T we see eclipses every month?

11. Easy memory rule: the eclipse is named after the object that gets _____.

ANSWER KEY

1) S 2) L 3) S 4) L 5) S 6) L 7) S 8) L | B: left = SOLAR (Moon in middle), right = LUNAR (Earth in middle) | 9) They must line up in a straight row | 10) The Moon's orbit is slightly tilted, so most months the shadow passes above or below | 11) blocked/darkened.

6 Seasons Myth-Buster

WORKSHEET

OBJECTIVE

Bust the "distance myth" and explain seasons using Earth's tilt and light angle.

TIME

20 minutes

STANDARDS

S6E2.c

Name: _____

Period: _____

Part A. Circle TRUE or FALSE. Then FIX every false statement on the line below it:

1. TRUE / FALSE — Summer happens because Earth moves closer to the Sun.

2. TRUE / FALSE — Earth's axis is tilted about 23.5 degrees.

3. TRUE / FALSE — The whole planet has summer at the same time.

4. TRUE / FALSE — Direct sunlight is more concentrated than slanted sunlight.

5. TRUE / FALSE — Earth is closest to the Sun in July, during northern summer.

Part B. Complete the table for the NORTHERN Hemisphere:

Month	Tilted toward or away from the Sun?	Sunlight direct or slanted?	Season
June			
December			

Part C. 6. When it's summer in Georgia, what season is it in Australia (Southern Hemisphere)? Explain WHY in one sentence.

7. If Earth had NO tilt at all, what would happen to the seasons? Explain.

ANSWER KEY

1) FALSE — seasons come from the tilt; Earth is actually closest in January 2) TRUE 3) FALSE — hemispheres have opposite seasons 4) TRUE 5) FALSE — closest in early January, during northern WINTER | B: June = toward / direct / summer; December = away / slanted / winter | 6) Winter — the Southern Hemisphere tilts away when the Northern tilts toward | 7) No seasons (or nearly none) — every place would get the same light angle all year.

7 Writing Prompts

WRITING

OBJECTIVE

Use scientific vocabulary to explain concepts in writing.

TIME

15–30 minutes each

STANDARDS

Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Dear Moon.** Write a short letter to the Moon asking it three questions about its phases — then answer one of the questions yourself using what you learned.
2. **One Day, One Year.** Explain the difference between Earth's rotation and Earth's revolution, and what each one causes. Use the words *axis* and *orbit*.
3. **Mirror in the Sky.** The Moon makes no light of its own. Explain where moonlight really comes from, and why we sometimes see only a sliver of it.

Level 2 — Intermediate (3 paragraphs)

4. **Eclipse Reporter.** You're a news reporter covering BOTH a solar eclipse and a lunar eclipse this year. Write two short news reports. Each must say which object is in the middle, whose shadow falls where, and what people on Earth see.
5. **The Upside-Down Vacation.** Your family is planning a December beach vacation to Australia. Your friend says, "December is winter — you'll freeze!" Explain why your friend is wrong, using Earth's tilt.
6. **Moon Myth-Buster.** Many people believe the Moon's phases are caused by Earth's shadow. Write an explanation that busts this myth, including what REALLY causes phases and when Earth's shadow actually does hit the Moon.

Level 3 — Advanced (essay)

7. **Design an Experiment.** A skeptical student doesn't believe that the angle of sunlight changes how much an area heats up. Design an experiment (materials, steps, what to measure) to convince them — then explain what results you'd expect and why.
8. **A World Without Tilt.** Imagine Earth's axis suddenly straightened to 0°. Write an essay describing how life would change: seasons, growing food, day length, animal migration. Support each prediction with the science of light angle.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE

Practice scientific thinking through verbal reasoning.

USE

Warm-ups, exit tickets, Socratic seminar

STANDARDS

Cross-cutting

TEACHER NOTES

These are designed for "think-pair-share" format. Give students 30 seconds to think, 1 minute to discuss with a partner, then call on a few groups. Some questions have no single "right" answer — push students to defend their thinking with evidence from the chapter.

Quick Warm-Ups (1–2 min each)

1. The Sun "rises" and "sets" every day. Is the Sun actually moving around us? What's really happening?
2. Is the Moon a planet, a star, or something else? How do you know?
3. If you lived on the Moon, would Earth have phases when you looked up at it?
4. Tonight there's a full moon. About how long until the next full moon? How do you know?
5. Why is it dangerous to look at a solar eclipse but safe to look at a lunar eclipse?

Deeper Discussions (5–15 min each)

6. The "Earth's shadow causes phases" myth feels very logical. Why do you think so many people believe it? What's the best single piece of evidence against it?
7. Earth is closest to the Sun in January — northern winter. If a friend refuses to believe distance doesn't cause seasons, what evidence would you show them first?
8. Some places near the equator barely have seasons, while places far north have extreme ones. Use the idea of light angle to explain why.
9. Ancient civilizations predicted eclipses centuries before telescopes. What would people need to observe and record over many years to pull that off?
10. If Earth rotated much more slowly — say one rotation per year — what would happen to day, night, and temperature where you live?

9 Moon Journal — One-Month Sky Watch

GROUP PROJECT

OBJECTIVE

Observe and record the real Moon for a month, then connect observations to the phase model.

TIME

4 weeks (5 min/night) + 1 class period

STANDARDS

S6E2.a

Project Description

Teams of 3–4 keep a shared Moon Journal for one month. Each night, ONE team member is "on duty" to observe the Moon and sketch it (cloudy night? note it and check a weather app's moon phase as backup). At the end of the month, the team assembles their sketches into a calendar poster and presents the pattern they found.

Journal Requirements (each night's entry)

- Date, time, and observer's name
- A sketch of the Moon's shape (shade the dark part)
- The phase name (best guess — the team can correct it later)
- One-line note: Where in the sky? Rising or setting? Right or left side lit?

Final Poster & Presentation

- All sketches arranged in calendar order
- Labels showing where the Moon was waxing and where it was waning
- The dates of the four "anchor" phases: new, first quarter, full, last quarter
- A 1–2 minute presentation: What pattern did you see? How long did one cycle take? Did anything surprise you?

TEACHER NOTES

Timing: Start the journal a few days before a new moon so students catch a full waxing cycle early.

Differentiation: Provide a pre-printed grid of empty circles for sketching; struggling students shade, advanced students also estimate % illuminated.

Reality check: Some nights the Moon rises after bedtime! Treat "I couldn't find it" as DATA — it leads to great discussion about moonrise times.

Grading rubric (suggested): Completeness of entries (40%), accuracy of phase labels (30%), poster quality (15%), presentation (15%).

10 Vocabulary Bingo

[REVIEW GAME](#)

OBJECTIVE

Review chapter vocabulary in a low-stress game format.

TIME

20–30 minutes

STANDARDS

S6E2 (all)

How to Play

1. Students write one word from the word bank into each square of their board, in any order. (FREE space stays free.)
2. The teacher reads a **definition** (not the word!). Students mark the square if they have the matching word.
3. First to 5 in a row calls "BINGO!" — but must read back each marked word AND its meaning to win.
4. Play again with "blackout" rules (whole board) for a longer review.

Word Bank (24 words + FREE)

rotation	full moon	solar eclipse	indirect sunlight
revolution	waxing	lunar eclipse	orbit
axis	waning	tilt	shadow
satellite	crescent	season	first quarter
phase	gibbous	hemisphere	last quarter
new moon	eclipse	direct sunlight	equator

		FREE 🌙		

TEACHER NOTES

Read definitions straight from the Vocabulary Reference sheet (Section 2) so wording matches what students studied. Keep a checklist of called words to verify BINGOs. For struggling readers, display the word bank on the board during play.